

ECON 108

INTRODUCTION TO ECONOMICS

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The macroeconomy: Unemployment, Inflation, and Deflation

CHAPTER V (from the book)

CHAPTER OUTLINE

Unemployment

Types of Unemployment

Inflation and Deflation

Business Cycles

UNEMPLOYMENT

What is unemployment?

- Total number of adults (aged 16 years or older) willing and able to work and who are actively looking for work but have not found a job.
- Unemployment creates a cost to the entire economy in terms of lost output – often ranging in the billions of dollars.

UNEMPLOYMENT

Labor Force

- Individuals aged 16 years or older who either have jobs or who are looking and available for jobs; the number of employed plus the number of unemployed.

UNEMPLOYMENT

Unemployment Rate

- The unemployment rate is the percentage of the measured **labor force** that is unemployed.

UNEMPLOYMENT

Cost of Unemployment:

Lost output

- During early 2000s, unemployment rate rose by 2 percentage points
- Firm output was 80% of potential
- Lost output was \$200 billion of goods and services that could have been produced

UNEMPLOYMENT

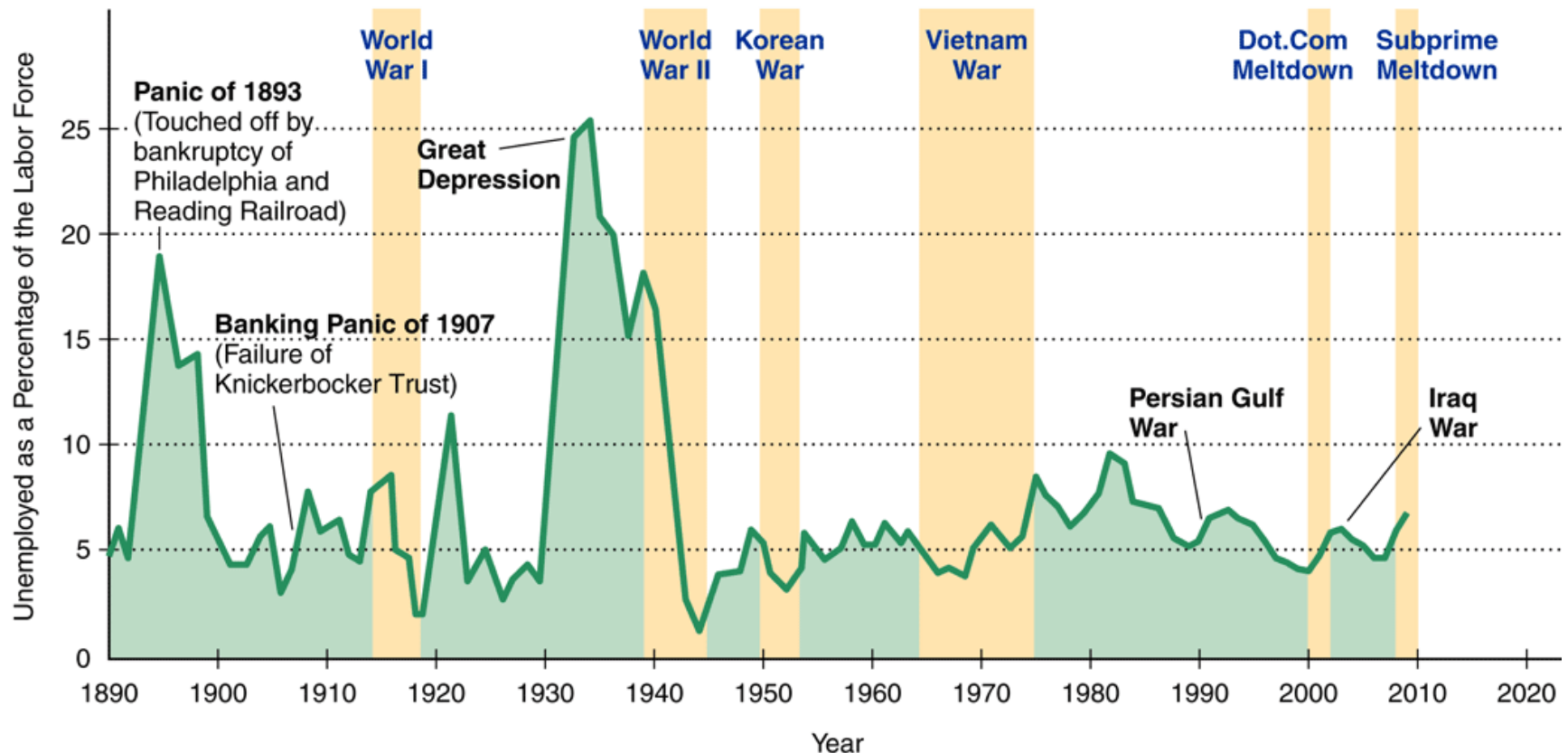
Cost of Unemployment (cont.)

Personal psychological impact

- Psychologists believe that being fired generates as much stress as the death of a close friend.

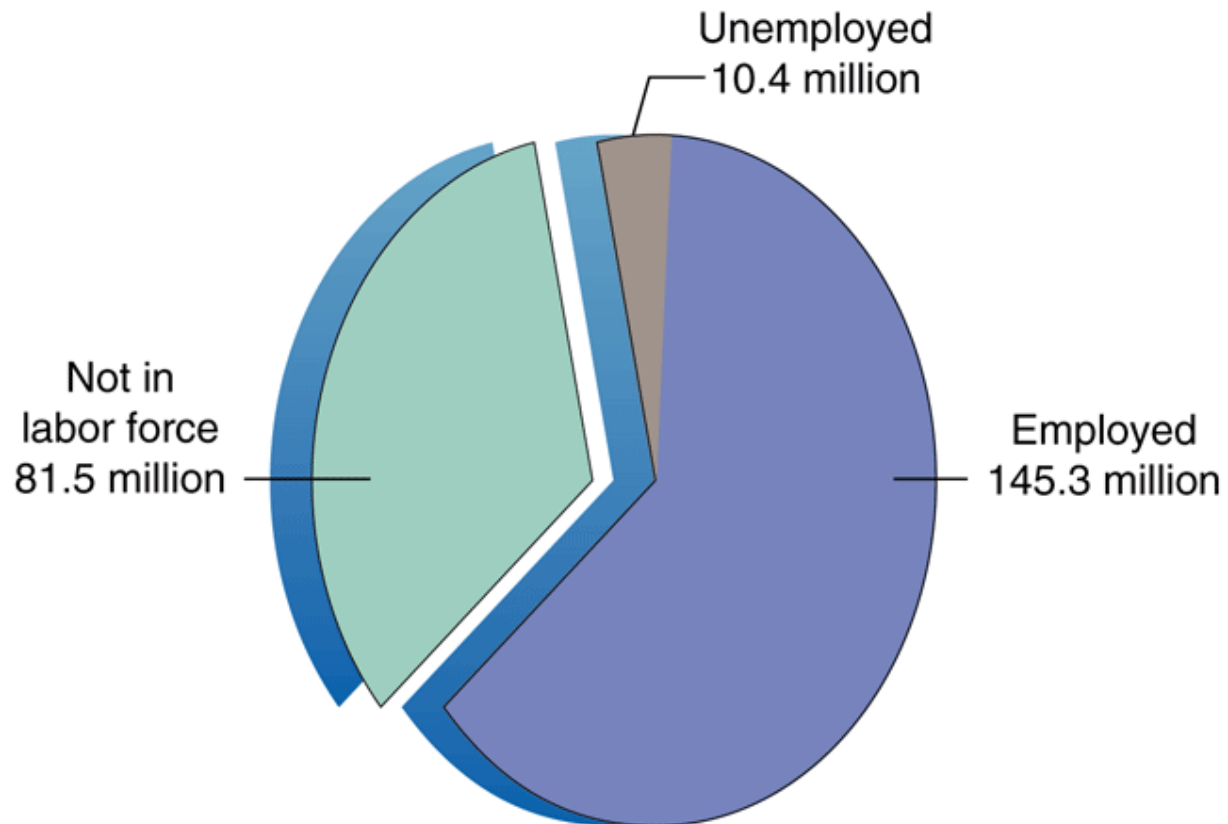
UNEMPLOYMENT

Historical Unemployment Rates



UNEMPLOYMENT

Employment, Unemployment, and the labor Force



UNEMPLOYMENT

The Arithmetic Determination of Unemployment

Labor force = The employed + The unemployed

$$155.7^* = 145.3 + 10.4$$

*U.S., millions of people; as of 2009

UNEMPLOYMENT

The Arithmetic Determination of Unemployment

$$\text{Unemployment rate} = \frac{\text{Unemployed}}{\text{Labor force}} \times 100$$

$$= \frac{10.4}{155.7} \times 100 = 6.7\%$$

*U.S., millions of people; as of 2009

UNEMPLOYMENT

Stock

- The quantity of something, measured at a given point in time—for example, an inventory of goods.

Flow

- A quantity measured over time, such as the income you make per year, or the number of individuals fired every month

UNEMPLOYMENT

Categories of individuals without jobs

- **Job loser:** individual whose employment was involuntarily terminated or who was laid off (40 to 60 percent of unemployed)
- **Reentrant:** individual who used to work full-time but has been out of the labor force (20 to 30 percent of unemployed)

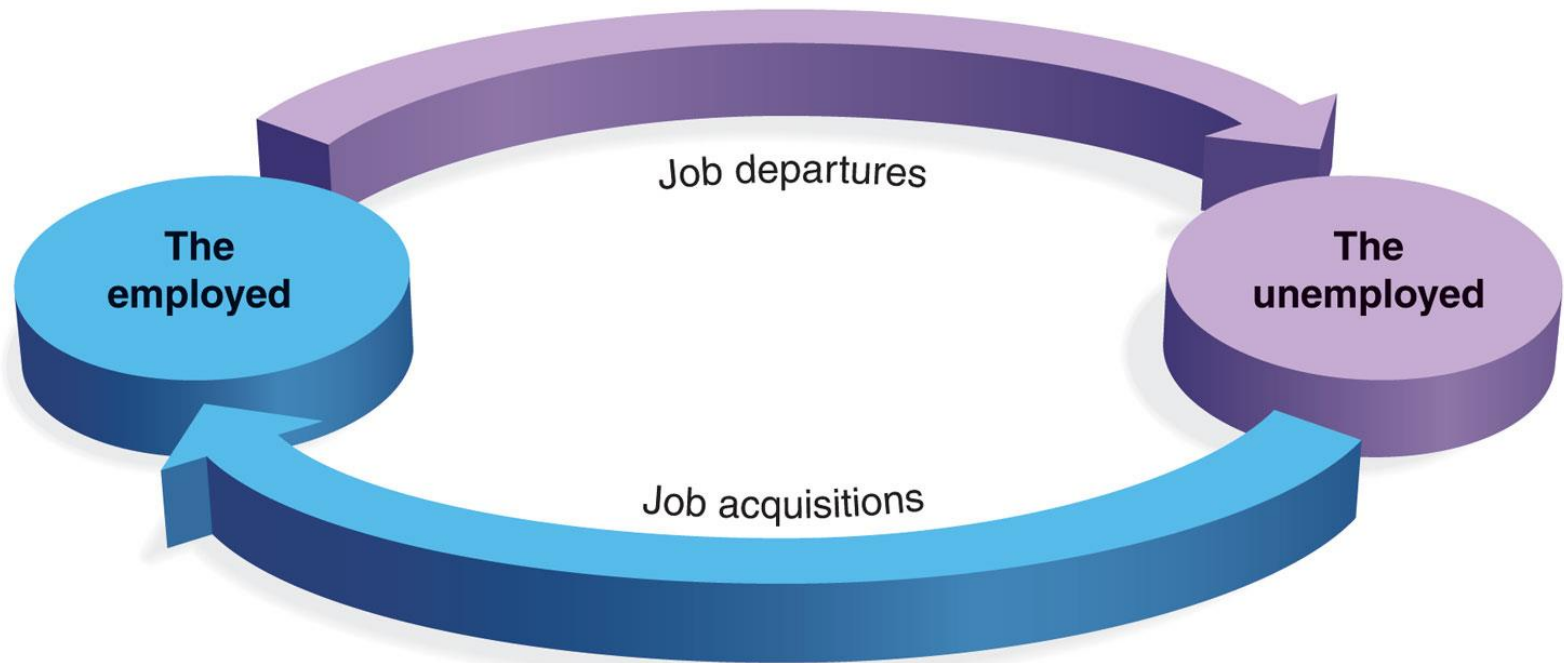
UNEMPLOYMENT

Categories of individuals without jobs

- **Job leaver:** individual who voluntarily ended employment (10 to 15 percent of the unemployed)
- **New entrant:** individual who has never worked full-time for two or more weeks (10 to 13 percent of the unemployed)

UNEMPLOYMENT

Logic of the unemployment rate



UNEMPLOYMENT

Duration of Unemployment

- More than a third of job seekers find work within one month.
- Approximately another third find employment within a second month.
- About a sixth are still unemployed after six months.
- Average duration is just over 15 weeks throughout the last 15 years.

UNEMPLOYMENT

Discouraged workers

- Individuals who have stopped looking for a job because they are convinced they will not find a suitable one.
- Discourage workers are not considered in the unemployment rate (hidden unemployment).

TYPES OF UNEMPLOYMENT

The mayor types of unemployment are:

- Frictional
- Structural
- Cyclical
- Seasonal

TYPES OF UNEMPLOYMENT

Frictional unemployment:

- Results from the fact that workers must search for appropriate job offers.
- This takes time, so they remain temporarily unemployed.

TYPES OF UNEMPLOYMENT

Structural unemployment:

- Results from a poor match of workers' abilities and skills with current requirements of employers.
- Government labor market policies influence how many jobs businesses wish to create, thereby affecting structural unemployment.

TYPES OF UNEMPLOYMENT

Cyclical unemployment:

- Results from business recessions that occur when aggregate (total) demand is insufficient to create full employment

TYPES OF UNEMPLOYMENT

Seasonal unemployment:

- Results from the seasonal pattern of work in specific industries.

Full Employment and the Natural Rate of Unemployment

Questions:

- Does full employment mean that everybody has a job?
- Is it always possible for everyone who is looking for a job to find one?

Full Employment and the Natural Rate of Unemployment

Full employment

- An arbitrary level of unemployment that corresponds to “normal” friction in the labor market.

Full Employment and the Natural Rate of Unemployment

Natural Rate of Unemployment

- The unemployment rate that is estimated to prevail in the long-run macroeconomic equilibrium.
- Should not reflect cyclical unemployment.
- When seasonally adjusted, the natural rate should include only frictional and structural unemployment.

INFLATION AND DEFLATION

Inflation

- A sustained increase in the average of all prices of goods and services in an economy

Deflation

- A sustained decrease in the average of all prices of goods and services in an economy

INFLATION AND DEFLATION

Purchasing Power

- The value of money for buying goods and services
- Varies with prices and income

INFLATION AND DEFLATION

Nominal Value

- Price expressed in today's dollars

Real Value

- Value expressed in purchasing power, adjusted for inflation, or in “constant dollars.”

INFLATION AND DEFLATION

Measuring the rate of inflation

- The **market basket** is a representative bundle of goods and services
- The **base year** is a point of reference (a specific year) chosen to compare prices to other years

INFLATION AND DEFLATION

Measuring the rate of inflation

- The **price index** is the cost of today's market basket of goods expressed as a percentage of the cost of the same market basket during a base year

$$\text{Price index} = \frac{\text{Cost of market basket today}}{\text{Cost of market basket in base year}} \times 100$$

INFLATION AND DEFLATION

Measuring the rate of inflation

Example:

(1) Commodity	(2) Market Basket Quantity	(3) 2001 Price per Unit	(4) Cost of Market Basket in 2001	(5) 2011 Price per Unit	(6) Cost of Market Basket in 2011
Corn	100 bushels	\$ 4	\$ 400	\$ 8	\$ 800
Computers	2	500	1,000	425	850
Totals			\$1,400		\$1,650
$\text{Price index} = \frac{\text{cost of market basket in 2011}}{\text{cost of market basket in base year 2001}} \times 100 = \frac{\$1,650}{\$1,400} \times 100 = 117.86$					

INFLATION AND DEFLATION

Real world price indexes

- Consumer Price Index (CPI)
- Producer Price Index (PPI)
- GDP deflator
- Personal Consumption Expenditure (PCE)

INFLATION AND DEFLATION

Consumer Price Index

- A statistical measure of a weighted average of prices of a specified set of goods and services purchased by wage earners in urban areas
- Market basket of goods and services of typical consumer

INFLATION AND DEFLATION

Producer Price Index

- A statistical measure of a weighted average of prices of goods and services that firms produce and sell
- Used as a short-run leading indicator (before CPI)
- PPIs for foodstuffs, intermediate goods, and finished goods

INFLATION AND DEFLATION

GDP deflator

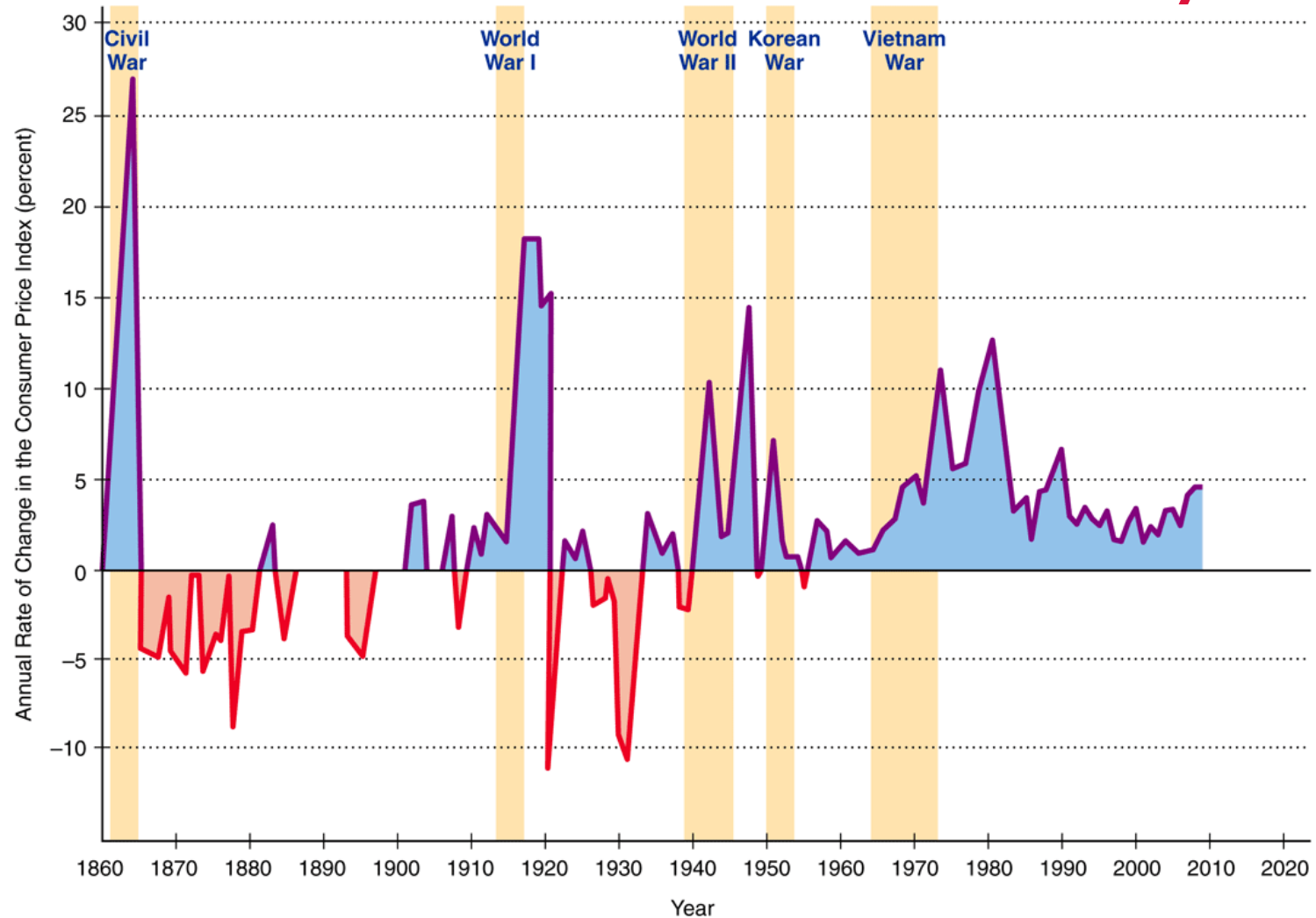
- A price index measuring the changes in prices of all new goods and services produced in the economy
- Broadest measure of prices; reflects both price changes and the public's market responses to those price changes

INFLATION AND DEFLATION

Personal Consumption Expenditure (PCE) Index

- A statistical measure of average price using annually updated weights based on consumer spending
- Primary inflation index used by the Federal Reserve

Inflation and Deflation in U.S. History



INFLATION AND DEFLATION

Anticipated versus unanticipated inflation

- The effects of inflation on individuals depend upon which type of inflation exists

Anticipated Inflation

- The inflation rate that we believe will occur

Unanticipated Inflation

- Inflation at a rate that comes as a surprise

INFLATION AND DEFLATION

Inflation and interest rates

Nominal Rate of Interest

- The market rate of interest expressed in today's dollars

Real Rate of Interest (expected)

- The nominal rate of interest minus the anticipated rate of inflation

INFLATION AND DEFLATION

Inflation and interest rates

Example:

- Nominal interest rate = 10%
- Expected inflation rate = 6%
- Real rate = $10\% - 6\% = 4\%$

INFLATION AND DEFLATION

Anticipated versus unanticipated inflation

Question: *Does inflation necessarily hurt everyone?*

- Inflation affects people differently. With unanticipated inflation creditors lose and debtors gain

INFLATION AND DEFLATION

Protecting against inflation

Cost-Of-Living Adjustments (COLAs)

- Clauses in contracts that allow for increases in specified nominal values to take account of changes in the cost of living

INFLATION AND DEFLATION

The resource cost of inflation

- Some economists believe that the main cost of inflation is the opportunity cost of resources used to protect against inflation. For example:
 - Consumers spend time and resources figuring out how to hedge against inflation (transaction costs)
 - Firms face costs associated with recalculating prices and printing new price lists (menu costs)

CHANGING INFLATION AND UNEMPLOYMENT: Business Fluctuations

Business fluctuations

- The ups and downs in business activity throughout the economy

CHANGING INFLATION AND UNEMPLOYMENT: Business Fluctuations

Business fluctuations

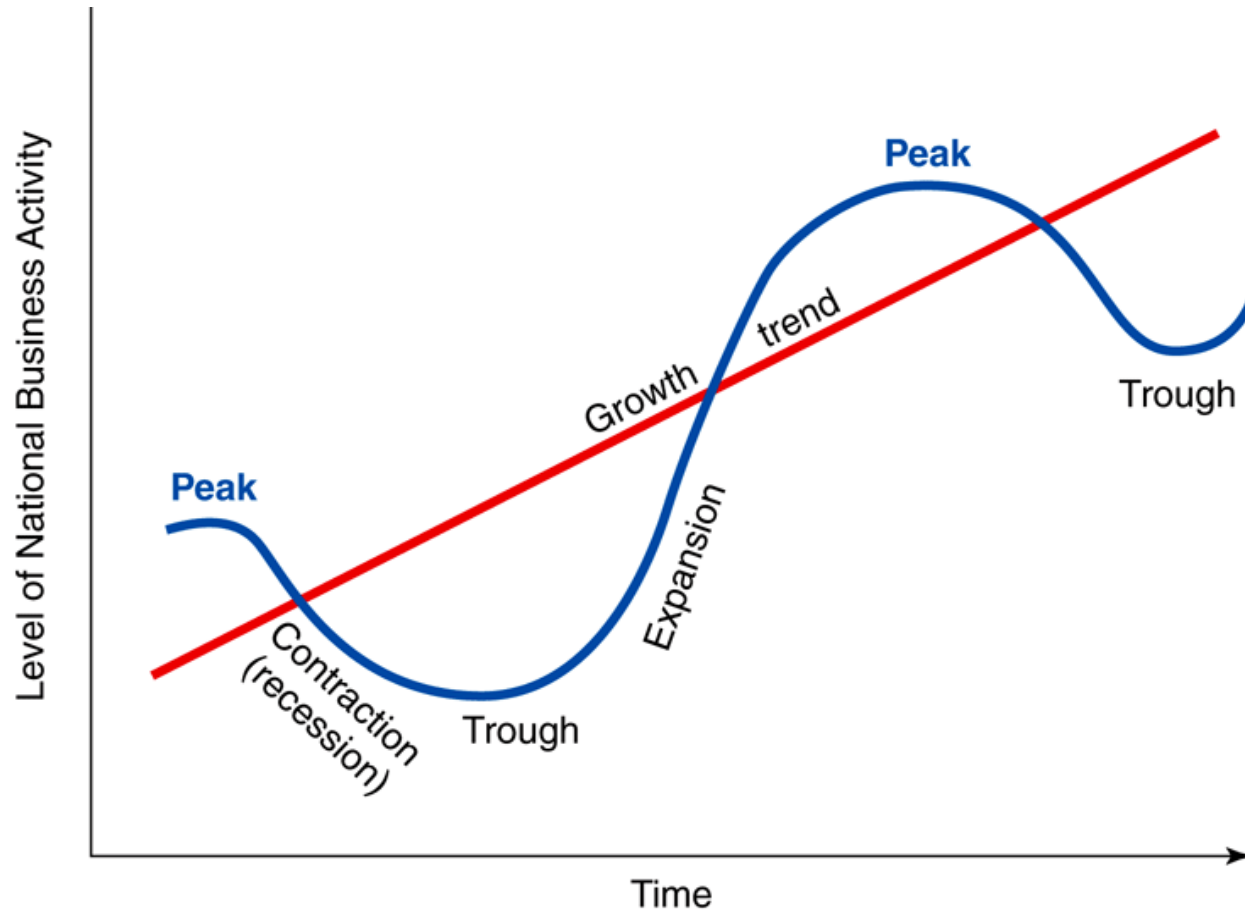
- An **expansion** is a business fluctuation in which the pace of national economic activity is speeding up
- A **contraction** is a business fluctuation during which the pace of national economic activity is slowing down

CHANGING INFLATION AND UNEMPLOYMENT: Business Fluctuations

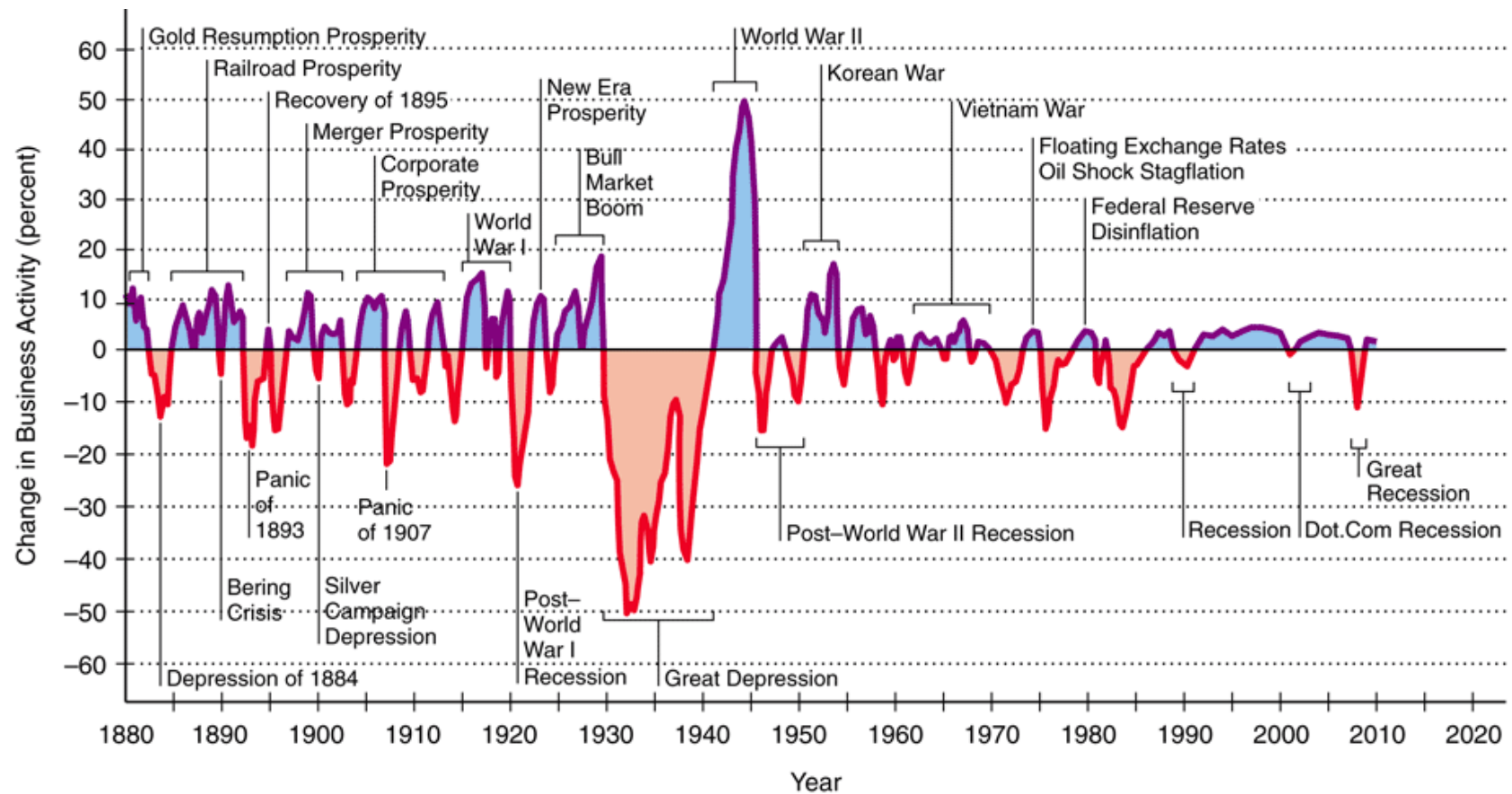
Business fluctuations

- A **recession** period of time during which the rate of growth of business activity is consistently less than its long-term trend or is negative
- A **depression** is an extremely severe recession

The Idealized Course of Business Fluctuations



National Business Activity, 1880 to the Present



Sources: *American Business Activity from 1790 to Today*, 67th ed., AmeriTrust Co., January 1996, plus author's estimates.

CHANGING INFLATION AND UNEMPLOYMENT: Business Fluctuations

Leading indicators

- Events that have been found to occur before changes in business activity. Economic downturns often follow:
 - Reduction in the average workweek
 - Rise in unemployment insurance claims
 - Decrease in prices of raw materials
 - Drop in the quantity of money circulating

END OF CHAPTER 5